

Scaling from Metawebs to Realised Webs: A Hierarchical Approach to Network Ecology

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Abstract: Ecological networks are increasingly used to understand biodiversity and predict ecosystem responses to environmental change, yet their application is often limited by uncertainty about the assumptions embedded in different network representations. Here, we present a theory-driven framework that views network construction as a hierarchical transition from metawebs, which represent the feasibility of interactions, to realised webs, which represent interactions expressed in specific spatiotemporal contexts. We identify five key processes underlying this transition: evolutionary compatibility and co-occurrence, which define interaction feasibility, and abundance, diet choice, and non-trophic interactions, which determine interaction realisation. We further position these processes along a continuum of network construction methods, from inductive approaches that infer structure from observations to deductive approaches that generate structure from mechanistic principles. By explicitly linking network representations to their underlying assumptions and scale-dependent processes, our framework clarifies the questions that different networks can address, highlights key challenges in moving between potential and realised interactions, and provides a roadmap for developing predictive network ecology. We argue that greater conceptual clarity is essential for advancing biodiversity science and for the effective application of networks in conservation, environmental management, and forecasting ecological change in the Anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

1 Introduction

At the heart of modern biodiversity science are a set of concepts and theories about species richness, stability, and function, which have been discussed since the foundational work of [1], [2], and [3] and more recently [4]. These relate to the abundance, distribution, functions, and services that biodiversity provides. Network representations of interactions among organisms are increasingly argued to be an asset to understanding and predicting the impacts of multiple, simultaneous stress on biodiversity (*e.g.*, foundational studies on food web structure and robustness: [5–8]). Documenting interactions is thus one of the fundamental building blocks of community ecology and provides a powerful abstraction for mathematical and statistical modelling of biodiversity to make predictions, and to mitigate and manage threats [9].

However, there is a growing discourse around limitations to the interpretation and applied use of networks, which have been recognised since early discussions of sampling effort and aggregation in food webs ([5,10,11]; recent discussions: [12,13]). Against this, it is important to evaluate the value and the limitations of the various network conceptualisations and how these relate to biodiversity concepts, such as community structure, community stability and ecosystem function [14]. In this perspective we aim to provide an overview of different **food web** representations, particularly how each representation embeds assumptions about the processes that determine interactions (Section 3) about the levels of organisation at which this occurs (*i.e.* the biological, ecological, spatial/temporal scales) and the way in which we construct the resulting networks (Section 4).

Network construction reflects both the data used and the theories governing species interactions. We still lack a clear explanation of the different assumptions and scale dependent processes that underpin network construction alongside extensive discussions about the challenges relating to data collection and observation [6,10,14–20]. Such an understanding should deliver an acceleration in capacity to more effectively predict the impact of multiple stressors on biodiverse communities and their functions.

In their recent work, [21] provided a perspective on the advantages and limitations of node resolution and link type. Here, we compliment this definition focused work by providing a theory-driven perspective that frames network construction as a hierarchical transition from feasibility to the realisation of interactions. In the following sections, we review how nodes, edges, and interaction processes shape different network representations, highlighting how changes in assumptions alter ecological node and link resolution, the

relative importance of evolutionary versus ecological processes, and how this intersects with the choices made when attempting to construct networks. We conclude by aligning these representations with key questions in biodiversity science in the Anthropocene.

2 Setting the Scene: The Not So Basics of Nodes and Edges

Ecological networks represent an ‘object’ from which inferences can be made and thus serve multiple purposes. While many aspects of community structure can be analysed without networks (*e.g.*, trait distributions or abundance patterns) networks provide a formal framework for capturing the organisation of interactions among species. The study of network structure and topology has a long history in ecology, rooted in early theory on energy flow [2,22], and later extended to questions of robustness, stability, and complexity (*e.g.*, [7,20,23–25]). More recent work has built on this foundation to link network structure to ecosystem functioning, persistence, and dynamical behaviour (*e.g.*, [26–28]). Networks are therefore commonly treated as response variables in tests of ecological theory and statistical models of the generative processes that give rise to observed structure and are widely used to compare communities across environmental gradients or through time [*e.g.*, 29,30]. They also provide a platform for evaluating downstream responses to perturbations, including secondary extinctions and robustness to species loss (*e.g.*, [7,31–33]), as well as inference about stability, ecosystem function, invasions, climate change, contaminants, and extinction cascades (*e.g.*, [34–36]). Against this backdrop of multiple research agendas, the definition of ‘edges’ and ‘nodes’, and the levels of organisation at which they are defined, take many forms [17,37], each of which encode a series of assumptions within a network. Here we introduce an alignment between these questions and baseline assumptions about nodes and edges.

2.1 How do we define a node?

While nodes are conventionally described as representing species, in practice they may correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera, or families, as well as trophic species (*e.g.*, [38,39]), feeding guilds (*e.g.*, [40]), or life-stage-specific subsets of species (*e.g.*, [41]). These choices reflect differences in the level, type, and consistency of resolution at which networks are constructed, rather than aggregation *per se*. Representing nodes at coarser or mixed resolutions can limit taxon-specific inference (*e.g.*,

whether species a consumes species b), bias estimates of degree distributions (particularly generality and vulnerability) and complicate downstream analyses such as extinction or invasion dynamics, where species identity and the consequences of loss may be obscured [41,42]. At the same time, there are clear justifications for using aggregated representations when the distribution of interactions among functional or trophic units is more informative than species-level detail, for example when analysing extinction patterns across feeding guilds or other functional units [43]. Issues of resolution, scale, and sampling have long been recognised as central to the construction and interpretation of food webs (*e.g.*, [6,44]) and it is important that we understand how the choices we make when defining nodes will impact the conclusions that are derived from a network [21].

2.2 What is captured by an edge?

Understanding how edge assumptions impact on ecological process representation requires distinguishing between *potential* and *realised* links - potential links reflect feasibility, whereas realised links reflect fluxes such as energy transfer. Links within food webs are thus a representation of either potential links between species [45] or fluxes within a system *e.g.*, energy transfer or material flow as the result of the feeding links between species [2,46]. Edges can thus correspond to different ‘currencies’ [21]. There is also a myriad of ways in which the links themselves can be specified. Links between species can be treated as present or absent (*i.e.*, binary), may be defined as probabilities [47,48] or by continuous functions which further quantify the strength of an interaction [49]. Link definition depends on both the ecological currency and how interactions are represented. For example, feasible links unlikely to accommodate inferences about the flow of energy through the network as the links are not environmentally/energetically constrained.

2.3 Network representations

These definitions of nodes and edges and associated assumptions give rise to two well established types of ecological networks: **metawebs**, which represent all *potential* interactions within a species pool [44], and **realised networks**, which represent the subset of interactions expressed within a particular community at a given place and time. Because of the assumptions made about nodes and edges in these networks, they differ in both their representation of scale at which they are constructed and the processes assumed to generate their structure.

96 A metaweb is fundamentally a list of *feasible* interactions between species. Metawebs
97 identify evolutionarily and regionally plausible interactions, ecologically impossible (*i.e.*,
98 forbidden) links [50], and the potential diet breadth of species [51]. Feasibility is thus
99 determined by trait complementarity, typically related to feeding, and can be further
100 constrained by species co-occurrence, producing a transition from *global* to *regional*
101 metawebs.

102 In contrast, realised networks are localised in space and time, with links shaped by co-
103 occurrence, environmental conditions, and diet choice. Even when represented as binary
104 matrices, their links implicitly reflect interaction strength and energy transfer. Realised
105 networks are therefore not simple spatial or temporal subsets of metawebs; they emerge
106 from the processes governing whether feasible interactions are expressed. Consequently,
107 a metaweb and realised network containing the same species may differ substantially in
108 structure because link presence is governed by different constraints. Critically, links absent
109 from a metaweb represent infeasible interactions, whereas links absent from a realised
110 network reflect both infeasible and context-dependent ecological constraints.

111 As a result of this distinction a realised interaction can have many definitions. At its most
112 general, a realised network requires more than the feasibility of species interactions; it
113 must incorporate the ecological context that determines whether feasible interactions are
114 expressed. This context may be represented through observation, abundance-dependent
115 encounter rates, consumer choice, environmental conditions, or other mechanisms that
116 constrain interaction expression. Consequently, realised networks need not be observed
117 directly, but they should reflect processes governing the distribution or strength of interac-
118 tions within a particular community. In contrast, metawebs represent the space of feasible
119 interactions irrespective of whether those interactions are expressed locally.

120 **3 From Nodes and Edges to Process and Constraints**

121 In the previous section we discussed how the definition of nodes and edges, representing
122 different scales and processes, lead to the concept of a metaweb and a realised web. The
123 fundamental take-homes are that nodes vary in their resolution, edges vary in what kind
124 of process they represent and the intersection of these, defined by meta- vs. realised webs,
125 underpins distinct lines of inquiry and constraints on the type of inference we can make
126 with networks.

127 Following this, we here reveal five core constraints across evolutionary and ecological scales
128 that further delineate the transition from meta- to realised webs, exposing processes that
129 determine the nature of links among nodes: evolutionary compatibility, co-occurrence,
130 abundance, diet choice, and non-trophic interactions Figure 1.

131 [Figure 1 about here.]

132 3.1 Processes that determine the feasibility of an interaction

133 Evolutionary compatibility and co-occurrence are the two principal processes that define
134 the feasibility of an interaction between two species. The scale of inference and set
135 of processes embodied in these two constraints typically combine to define a ‘list’ of
136 interactions that are viable/feasible and defined strictly as present/absent. Reflecting on
137 the previous section, nodes are typically species and rules defining edges are defined by
138 trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight
139 into each process.

140 Evolutionary compatibility

141 This constraint is defined by shared (co)evolutionary history between consumers and
142 resources [52–55] which is manifested as ‘trait complementarity’ between two species [56].
143 In this body of theory, the consumer has the ‘correct’ set of traits that allow it to acquire
144 and consume the resource. Interactions that are not compatible are defined as forbidden
145 links [50]; *i.e.*, they are not physically possible and will *always* be absent within a network.

146 Networks do not explicitly arise from models based on this constraint. Instead, interacting
147 species pairs are defined and these are represented as binary (possible vs forbidden) or
148 probabilistic [47]. For example, in the metaweb constructed by [57] probabilities are
149 quantified as the confidence of a specific interaction being *possible* between two species. A
150 network constructed based on evolutionary compatibility is conceptually aligned with a
151 ‘global metaweb’, and gives us information as to the global feasibility of links between
152 species pairs even though they do not co-occur (see Figure 1).

153 (Co)occurrence

154 The co-occurrence of species in both time and space is a fundamental requirement for an
155 interaction between two species to occur (at least in terms of feeding links). Although
156 co-occurrence data alone is insufficient for building an accurate and ecologically meaningful
157 representation of *feeding links* [58], it is still a critical process that determines the possible

158 realisation of a feeding. Knowledge on the co-occurrence of species allows us to spatially
159 constrain a global metaweb to reflect regional metawebs [59]. In the context of Figure 1
160 this would be the metawebs for regions one and two.

161 We reinforce that these two constraints don't deliver a network *per se*, but a list of feasible
162 species pairs. Although it is possible to visualise a network from the list of interactions
163 generated by these constraints, it is important to be aware that the structure of this
164 network is not constrained by any community context - just because species are able to
165 interact does not mean that they will [60,61].

166 3.2 Processes that realise networks

167 In contrast to the above, here we highlight three processes that influence the *realisation* of
168 an interaction between species and thus form the conceptual basis for realised networks.
169 As we show in Figure 1, a 'truly realised' network is the product of properties of the
170 community (**abundance** and **non-trophic interactions**) and the individual (**diet**
171 **choice**). This represents a conceptual shift from considering the feasibility for species
172 pairwise interactions to considering the edge as a representation of energy flow. Such
173 a transition requires information about how the community, the environment and the
174 individual *constrains* network topology as defined by consumer choice ([62], Section 2.3)

175 Abundance

176 Abundance as a realising process emerges from a null model for energy acquisition:
177 organisms feeding randomly will consume resources in proportion to their abundance
178 [63]. Here, abundance of different prey species influences the distribution of links in a
179 network [64] by defining a preference linked to individuals among species meeting [47,60].
180 Abundance data (linked to a derived metaweb) delivers a foundation ruleset that can
181 define the distribution and strength of links. Of note, however, is that such abundance
182 constrained interactions are not necessarily contingent on there being any compatibility
183 between species [65–67].

184 Diet choice

185 It is well established that consumers make more active decisions than eating items in
186 proportion to their abundance [63]. Ultimately, consumer choice is underpinned by
187 an energetic cost-benefit framework centred around profitability and defined by traits
188 associated with acquisition and consumption of a resource [68,69]. Energetic constraints

189 are invoked to construct networks in a myriad of ways [*e.g.*, 42,70–72].

190 Unlike metaweb approaches, these models generate realised webs as emergent outcomes of
191 consumer behaviour. We also here make a distinction, developed below, with models like
192 the Niche Model [39], where diet choice is implicit in its probabilistic network generating
193 function, but it is working to replicate the *expected* structure of the network, and this
194 structure does not emerge from node-based rules. Note that we select diet choice as a term
195 to capture rules linked to optimal foraging [73] and metabolic theory [74] for capturing
196 the energetic constraints on the distribution and strength of interactions.

197 **Non-trophic interactions**

198 We include non-trophic interactions [see 75] here not as a determinant of links, but a
199 modifier of them - they are the community context above and beyond co-occurrence and
200 abundance. Non-trophic interactions include competition for space, predator interference,
201 refuge provisioning, recruitment facilitation as well as non-trophic effects that increase or
202 decrease mortality. These interactions specifically modify either the realisation or strength
203 of trophic interactions [28,31,76–78] and represent direct (*e.g.*, predator *a* outcompetes
204 predator *b*) and indirect (*e.g.*, mutualistic/facilitative interactions) mechanisms.

205 Some interactions, such as pollination, occupy an intermediate position in this framework,
206 as they combine trophic components (*e.g.*, resource consumption) with non-trophic effects
207 that influence reproduction, recruitment, and population persistence [79–81]. They operate
208 on the realisation of a network by altering the fine-scale distribution and abundance of
209 species and relative contributions of direct and indirect effects to biomass, persistence,
210 stability, and the functioning of the communities [75,82–84].

211 **4 Network construction**

212 The above five processes are central to understanding the assumptions inherent in building
213 different types of networks. Each of the processes, or combinations thereof, deliver
214 a unique set of boundary conditions on what a network represents and can be used
215 for. Here we build on the introduction of these five processes to further categorise
216 the approaches to constructing networks ultimately moving to showcase how different
217 construction approaches, encoding different assumptions about how interactions arise, can
218 support different ecological questions.

4.1 Why construct networks?

Networks are a representation of biodiversity. In a perfect world, we might know about all interactions. However, the empirical collection of interaction data is both costly and challenging to execute [50,85,86]. In the absence of robust empirical data, we use ‘models’ that facilitate interpolation and gap-filling of existing empirical datasets [*e.g.*, 87,88–90], predict the feasibility of interaction among pairs of species, or directly predict network structure [see 91 for a broader discussion].

Networks are unique in delivering more than just estimates of species richness. As noted in the introduction, a network embodies the organising structure of biodiversity and allows numerous opportunities for ‘downstream’ analysis, including the comparison of structures, estimation of energy flux or extinction dynamics and ultimately form the structural inputs to dynamical systems models that facilitate ecological and conservation relevant inference about productivity-diversity-stability-function relationships [27] in space and time. But making such inferences requires careful attention to one or more of the processes discussed in Section 3. While these network representations may be simplifications of the truth [92], they remain critically useful tools as they allow us to move beyond simple descriptions of species richness to test hypotheses about community architecture and ecosystem functioning that would be otherwise impossible to assess.

4.2 Construction through induction

Constructing feasible or realised networks can be framed as an ‘inductive reasoning’ process where insight and generalisation arise from a set of observations and relationships around feeding. Inductive reasoning as a foundation for network construction is implemented at node and network levels.

4.2.1 Species specific induction

Species-specific approaches construct networks from the expected feeding interactions between species pairs and are therefore fundamentally creating metawebs. All methods in this inference space rest on a set of three assumptions: there are a set of ‘feeding rules’ that underpin interaction feasibility [93]; these rules are phylogenetically conserved [54,94]; and they can be specified by matching the traits between consumer and resource.

Evolutionary compatibility and co-occurrence constraints have been critical to the construction of ‘first draft’ networks for communities for which we have no interaction data [57].

They are also central to interpolation in data poor regions and predicting interactions for ‘unobservable’ communities *e.g.*, prehistoric networks [43,95,96] or future, novel community assemblages [97]. Furthermore, they have the capacity to evaluate a role of interactions among species relative to their distribution by accounting for the role of the environment and the role of species interactions [98–101].

Feeding rules can be specified in several ways. Expert-based approaches define feasible interactions using trait matching [102,103], while mechanistic approaches often rely on trait-based relationships between consumers and resources [104,105]. Alternatively, statistical and machine-learning methods infer interaction probabilities from observed networks using ecological predictors such as traits and phylogeny [106].

Rules are also defined by correlating real world interaction data with suitable ecological proxies for which data is more widely available (*e.g.*, traits) using some sort of binary classifier (see [106] for an overview). These include generalised linear models [*e.g.*, 107], random forest [*e.g.*, 108], trait-based k-NN [*e.g.*, 109], and Bayesian models [*e.g.*, 110,111].

Finally, graph embedding uses the structural features of a known network to infer the position of species in an unknown network through the decomposition of the interaction onto the embedding space (see [51]). This decomposition relies on a combination of ecological proxies (*e.g.*, phylogenetic relatedness [57]) in conjunction with known interactions to infer the latent values of species, which can then be mapped onto decomposition of a known network.

4.2.2 Structure-based induction

These models generate ecologically realistic network structures using simple probabilistic rules. They are commonly used as null expectations, for comparative analyses, and as inputs to dynamical models. The determination of links between species is not directly linked to properties of the nodes. This means these networks are usually not species specific. Although they require little empirical information, they encode explicit assumptions about expected network structure.

Stochastic network models use a probabilistic ruleset about diet choice and niche breadth to reflect fundamental ideas of foraging biology. These models that are based on the compartmentalisation and acquisition of energy for species at different trophic levels [112,113] and that network structure can be determined by distributing interactions along single dimension (the ‘niche axis’; [114]). Typically, these models parametrise some aspect

of the network structure (although see [112] for a parameter-free model). These models include the most used network generator, the Niche model [39], as well as the original Cascade model [5] and the derived Nested hierarchy model [115]. Even though these networks are derived without any real-world data they are still able to recover the structure of empirical networks [116]. These models often form the basis for dynamic models *e.g.*, the allometric trophic network [24,26] and bioenergetic food web models [34].

4.3 Construction through deduction

In contrast to metaweb construction, realised networks require assumptions about the ecological processes that determine whether feasible interactions are expressed. These approaches are deductive because interactions emerge from explicit assumptions about encounters, foraging behaviour, energetic constraints, or interaction modification. These approaches operationalise the abundance, diet-choice, and non-trophic processes introduced in Section 3.1. The resulting networks are widely used to study energy flux, extinction dynamics, and ecosystem functioning. They also provide the structural backbone for dynamical systems modelling to address questions about stability-structure-productivity-function relationships, secondary extinction dynamics, species invasion and climate change. There are two broad groups of models in this deductive category.

These models capture the *behaviour* of the nodes by explicitly considering the properties of the different species in the community. Which means that there is a degree of variance in which links are predicted between species unlike the more ‘static’ predictions made by inductive models. However, these networks are costly to construct in real world settings (requiring data about the entire community, as it is the behaviour of the system that determines the behaviour of the part) and lack the larger diet niche context afforded by metawebs.

4.3.1 Abundance-based models

Neutral networks are built on the assumption that foraging decisions are tied *only* to the abundance of species within the community [117,118]. Here links are solely determined by the relative abundance of the different species in the community. Although unrealistic as a complete explanation, neutral models can be combined with inductive approaches to generate more localised predictions [67].

312 4.3.2 Energetics-constrained models

313 There is a broader group of models that focus on determining interactions in terms of
314 energetic constraints on diet breadth, often using the ratio of consumer-resource body
315 size as a proxy for capturing the energetic constraints of feeding. Models such as those
316 developed by [71] and [68] are similar to the mechanistic approaches discussed in Section 4.2,
317 however instead of determining interactions based on mechanistic feasibility it is rather
318 constrained by the energetic cost of predation. Note that although these models do
319 not place any explicit constraints on the expected structure of the network, the links
320 should still be considered as ‘realised’ owing to the energetic constraint placed on links. A
321 different subset of diet models [*e.g.*, 42,119] use a diet choice approach, however like the
322 stochastic network models they also embed assumptions on network structure. Thus, these
323 models predict both interactions and network structure simultaneously, although they
324 may benefit in being refined by more explicitly accounting for trait-based (*i.e.*, feasibility)
325 parameterisation [35].

326 5 Making Progress with Networks

327 The motivation to leverage network ecology in conservation ecology, environmental risk
328 assessment and natural resource management stems from a shift away from species/popu-
329 lation specific measures of the effects of stress and disturbance to community level metrics
330 of these impacts. These metrics, such as resilience and more generally stability, ecosystem
331 function and biodiversity, are natural properties of networks. This suggests that modern
332 conservation, risk assessment and resource management would benefit from robust network
333 tools to support decision making.

334 This is also true in the disciplines of ecology and environmental science and their focus on
335 abundance, distribution, functions and services that biodiversity provides [4]. Major ques-
336 tions remain unresolved, for example, about stability-diversity-productivity relationships,
337 the drivers and consequences of extinctions and invasions, and the impacts of multiple
338 stressors operating at multiple ecological scales [8]. A network approach to answering
339 these types of questions specifically allows us to evaluate how environmental gradients and
340 anthropogenic stress map through direct and indirect effects among species in a complex
341 community and reveal fundamental patterns and understanding of processes in the natural
342 world.

343 To effectively use networks to aid us in answering questions about conservation/risk
 344 assessment/management and core ecological theory, we need to be mindful that we are
 345 mapping the *correct* network representation to the question of interest [21]. Notably, there
 346 are certain questions that cannot be answered using specific network representations as
 347 the scale of the question of interest is fundamentally misaligned with the process captured
 348 by a specific network representation (Section 3.1), the underlying data that is used to
 349 construct it (Section 4), or both.

350 Here we discuss and map the different network representations shown in Figure 1 to
 351 ‘appropriate’ research questions and agendas (see also Table 1). We also highlight some of
 352 the key methodological challenges that currently limit our conceptualisation of a ‘network’
 353 and thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally, we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $\checkmark$ indicates area with strong foundational research, Δ partial/emerging areas of research, and $\times$ areas where research is weak/largely absent

Network		
Representation	Example Research Question	Representative Studies
Global Metaweb	How will novel communities respond to <i>e.g.</i> , extinction, turnover, invasion and rewilding $\checkmark$	[120]; [7]
	Diet-based conservation focusing not only on the target species but the species it might depend on for food resources Δ	[121]; [122]; [123]
	Rewiring capacity/potential of species by looking at the <i>entire</i> diets of species Δ	[124]; [31]; [125]; [126]
	Eco-Evolutionary dynamics and how they relate to the conservation and origination of feeding strategies $\times$	[60]; [127]

Network		
Representation	Example Research Question	Representative Studies
Regional Metawebs	Applied use potential of questions highlighted for global metawebs at the management scale <i>e.g.</i> , a protected area Δ	[101]; [128]; [129]
	Refinement/extension of species distribution models by incorporating co-occurrence and species associations <i>e.g.</i> , predator and prey $\checkmark$	[130]; [131]
Realised Webs	The allocation of multiple stressors across networks Δ	[132]; [133]
	Temperature threshold to community collapse Δ	[134]; [135]
	Extinction and persistence after harvesting/invasion/extinction $\checkmark$	[136]; [137]
	Stability-diversity-productivity-function $\checkmark$	[138]; [121]
	Explicitly tying ecosystem level processes and nutrient flows to networks $\times$	[139]
	Meta communities and the idea of meta-network-communities Δ	[140]; [141]

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

Global metawebs are most appropriate for questions centred on interaction feasibility and potential diet breadth. They provide a platform for exploring hypothetical or novel communities under climate change [142], species invasions, reintroductions, and rewilding,

and the potential rewiring capacity [126]. Because they focus on feasible rather than realised interactions, global metawebs are also well suited to studying eco-evolutionary dynamics and how evolutionary history, natural selection, and phenotypic plasticity shape interaction niches.

5.1.2 Regional Metawebs

Regional metawebs extend these questions by explicitly incorporating species co-occurrence and therefore provide a more management-relevant perspective. They can be used to refine species distribution models and projections of future community composition [29,40]. However, caution is required when comparing regional metaweb structure across space or environmental gradients, as observed differences may reflect species turnover (*e.g.*, β -diversity) rather than changes in interaction processes.

5.1.3 Realised networks

Realised networks are best suited to questions concerning how community and environmental context shape interaction structure and ecosystem functioning [143]. They provide the appropriate framework for studying stability, resilience, biodiversity dynamics, ecosystem functioning, extinction cascades, invasions, climate change impacts, and network rewiring through time. By capturing interactions that are actually occurring, realised networks allow investigation of how perturbations propagate through communities and influence persistence.

The increasing availability of long-term interaction datasets is expanding opportunities to address these questions [27,144]. However, empirical datasets often accumulate interactions across extended periods, potentially obscuring temporal variation in realised interactions [10]. Developing approaches that better reconcile empirical networks with realised community dynamics therefore remains an important challenge.

5.2 Key methodological challenges

As noted above, the three network types highlight longstanding methodological challenges that limit both the precision and accuracy of ecological inference. Here we briefly review these challenges and emerging opportunities to address them.

Understanding what empirical data represents: Robust inference requires understanding what constitutes an observed interaction, whether recorded directly (predation

389 events) or indirectly (*e.g.*, gut contents or stable isotopes). Because empirical networks
390 often accumulate observations across space and time, they may be conceptually closer to
391 metawebs than realised networks.

392 **The validation of network structure:** Considerable progress has been made in
393 assessing how well models recover pairwise interactions [91,145], yet there remains no clear
394 framework for evaluating their ability to recover network structure [114,146]. This raises
395 two related questions: what constitutes an appropriate benchmark, and which aspects of
396 network recovery matter most? For metawebs, accurately identifying both present and
397 forbidden links may be essential, whereas for realised webs it remains unclear whether
398 recovering pairwise interactions, aggregate properties (*e.g.*, connectance), or both should
399 be the primary objective.

400 **Transitioning between metawebs and realised webs:** Most approaches for modelling
401 realised networks do not explicitly incorporate evolutionary constraints (although see
402 [147]; [68]). Progress will likely require either ensemble approaches [148,149] or methods
403 for downscaling metawebs into realised networks [*e.g.*, 150]. However, any such transition
404 must retain clarity about the meaning of links - structurally realistic networks do not
405 necessarily represent realised prey choice or energy flow.

406 Developing frameworks that allow transitioning between metawebs and realised rep-
407 resentations will also facilitate integration with metacommunity, metaecosystem, and
408 ecosystem-level theory [151,152]. Doing so requires that we develop an understanding of
409 what the appropriate spatial and temporal scales of networks are [153], and specifically
410 how these relate to network type and the question of interest.

411 **Making networks more tractable in applied spaces:** The application of networks
412 to conservation and management remains limited by difficulties in defining network
413 boundaries, aligning them with management units, and interpreting network metrics in
414 policy-relevant ways [154]. Addressing these challenges will require stronger links between
415 network structure and ecosystem function, alongside careful matching of analytical tools
416 and network representations to management objectives [128,155].

417 Taken together, these challenges highlight three overarching messages. (i) Network
418 representations are inseparable from the data and assumptions used to construct them;
419 (ii) validation and benchmarking must be aligned with the intended network type and
420 research question; and (iii) greater conceptual clarity is needed when transitioning between
421 metawebs and realised networks, particularly in applied contexts. Explicitly articulating

these distinctions is essential if networks are to be used rigorously and transparently across scales.

6 Concluding Remarks

1. **Network representations encode different processes:** Different network representations capture distinct assumptions, data structures, and ecological processes; understanding the interplay between a network's structure and the processes it encodes is critical for determining which questions it can address.
2. **Network representations are not interchangeable:** There is no universally optimal network representation. The utility of a network depends entirely on how well its underlying assumptions and data align with the intended scale and process of analysis.
3. **Explicit assumptions are essential for evaluation:** Explicitly articulating the assumptions behind network assembly facilitates a more critical evaluation of network suitability and inferential power. It is important that we move beyond the uncritical and default reliance on conventional network representations in favour of understanding the consequences of selecting a specific representation or reconstruction/modelling approach.
4. **A unified framework improves application and progress:** The framework presented here provides a structured basis for comparing network representations and evaluating their suitability across conceptual, methodological, and applied contexts. Establishing this standard is essential for preventing the misinterpretation of network data and for addressing foundational questions in biodiversity science.

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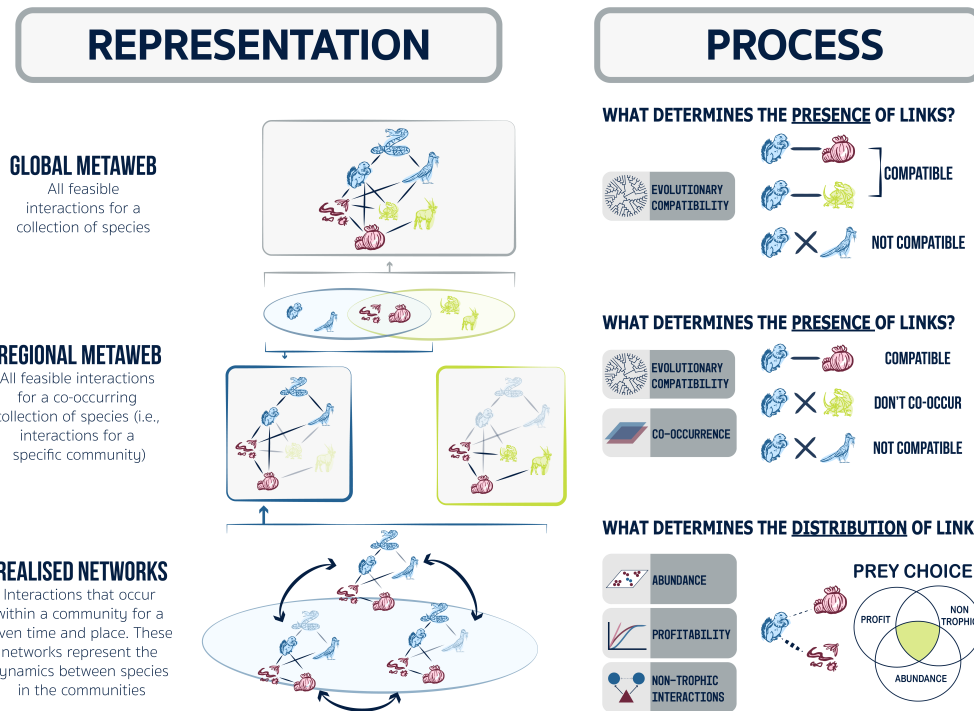


Figure 1: Aligning the processes that determine interactions (right) with different network representations (left). A **global metaweb** captures all feasible interactions among a species pool. Restricting this network by species co-occurrence yields **regional metawebs**, which differ in species composition. Species occurring in both regions are shown in red, whereas region-specific species are shown in blue and yellow. **Realised networks** represent subsets of regional metawebs expressed in specific spatial and temporal contexts. Their structure is shaped not only by co-occurrence, but also by community-level processes such as abundance, diet choice, and non-trophic interactions.